HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **low-latency C++17 data ingestion pipelines** for real-time market data feeds, processing over **50,000 messages per second** during peak trading hours.
- Developed and maintained **performance-critical C++ components** for trade execution and risk management systems, focusing on object lifetimes and memory management to achieve **sub-millisecond latency**.
- Refactored and optimized existing C++ codebases, identifying and resolving performance bottlenecks that reduced average execution time by **30%** for core trading algorithms.
- Contributed to the design and implementation of **system-wide data lifecycle management**, ensuring data integrity and efficient resource utilization across ingestion, processing, and output layers.
- Participated in rigorous **code reviews**, providing constructive feedback and ensuring adherence to coding standards for a team of 6 engineers.
- Collaborated with quantitative researchers and traders to understand requirements and translate them into **robust and maintainable C++ solutions**, participating in sprint planning and technical estimation.
- Containerized C++ trading applications using **Docker** and deployed them to a **Kubernetes** cluster on AWS, managing resource allocation and ensuring high availability.

SKILLS

Languages – C++, C++17, C++20, Python, Java, SQL

Systems – High-Throughput Systems, Performance-Sensitive Systems, Linux, Real-Time Systems, Distributed Systems

Tools & Technologies – Docker, Kubernetes, AWS, Git, Jenkins, Jira, Confluence, Splunk, Dynatrace

Fundamentals – Object Lifetimes, Memory Management, Resource Management, Data Lifecycle, Design Patterns, Agile Methodologies

Databases – PostgreSQL, MySQL, Oracle

Scripting & Automation – Python Scripting, Shell Scripting

PROJECTS

Low-Latency Market Data Ingestion Service (HCL Technologies)

- Engineered a high-throughput C++ service to ingest and parse real-time market data streams from multiple exchanges, achieving a sustained rate of **45,000 messages per second**.
- Implemented efficient memory management techniques, including custom allocators, to minimize garbage collection pauses and ensure consistent **sub-500 microsecond processing latency**.
- Integrated the service with a **PostgreSQL** database for historical data storage and analysis, developing SQL schemas to support time-series queries.

Performance Optimization for Trading Algorithm (MSc Project)

- Analyzed and profiled an existing C++ trading algorithm, identifying CPU-bound operations and memory allocation inefficiencies.
- Refactored critical code paths using modern C++ features and optimized data structures, resulting in a **28% reduction in execution time** for backtesting simulations.
- Developed unit tests using **Google Test** to verify the correctness of optimized code segments and ensure no regressions were introduced.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)